

Discussion paper for CIE review of the eastern Bering Sea walleye pollock stock assessment

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Executive Summary

The work presented below was based on the same data sets used in the 2024 assessment. The first task was to update the model runs to include the 2023 data. The model runs were updated and the results are presented below. The model runs were then used to compare the FT-NIR data with the traditional data sources (BTS and ATS). The results of these comparisons are presented below. The model fits to the data are presented in the tables at the end of the document. The first section steers towards the R library used for processing model results for EBS pollock. The second section presents the comparisons of the FT-NIR data with the traditional data sources. The third section compares results of different software platforms (e.g., SS3, CEA, and AMAK) for the assessment. The fourth section shows results comparing newer model-based estimates from survey data.

R Library for assessment compilations (ebswp)

The **ebswp** library is a collection of functions for compiling and analyzing results from the stock assessment models for the eastern Bering Sea walleye pollock (Ianelli (2025)). The library is designed to work with the ADMB code and depends on a consistent format for the report files generated. The link in the reference provides access to some notes on applications as vignettes to the package (e.g., . [here](#)).

Comparing FT-NIR compositions with traditional microscope age estimations

The fish age and growth lab at the AFSC have endeavored to expand on new technologies (Helser et al. (2019)).

FT-NIRS data consistency comparisons

Examining the abundance-at-age (for design-based estimates of the bottom-trawl age compositions) we show that in general, the TMA estimates show a high level of consistency by cohort (Figure 1). For these same data, when applied to the FTNIRS age-estimation method, the consistency was lower (Figure 2). The extent that this is a concern is unknown.

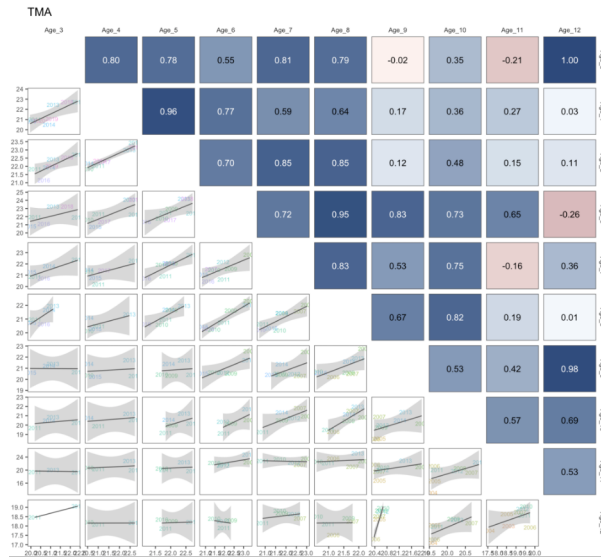


Figure 1: Cohort consistency over different ages based on log-abundance from the bottom-trawl survey data for pollock in recent years for traditional microscope age estimates. The cohorts are shown in the labels.

When preparing the data for the assessment, we plotted out the proportions-at-age for the TMA data and noted the differences in the estimates using the FTNIRS age estimates (Figure 3,

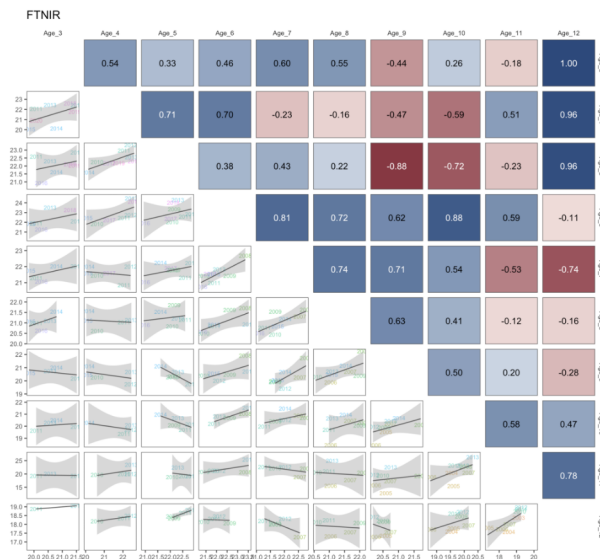


Figure 2: Cohort consistency over different ages based on log-abundance from the bottom-trawl survey data for pollock in recent years for FTNIRs age estimates. The cohorts are shown in the labels.

Figure 4, and Figure 5). Other preparations involved estimating the age-determination errors for each method following that of Punt et al. (2008). For the FTNIRs data, we optionally estimated separate age-estimation error matrices by gear type. The appearance of the fleet-aggregated and fleet-specific input data files is shown in Figure 6.

For the assessment model, the alternatives are shown based on the coverage of data and availability among types of ageing is shown in **Figure 7**. These figures are intended to show what data are available for the model testing and the different ways of treating age determination error matrices.

Comparisons of fit to FT-NIRS models

The layout of model runs comparing the traditional microscope ages (TMA) with the FT-NIRS ages is shown in Table 1. The models are run with the FT-NIRS data with different age-error matrix specifications. The model fits to the data are shown in Table 2. The table shows the goodness-of-fit

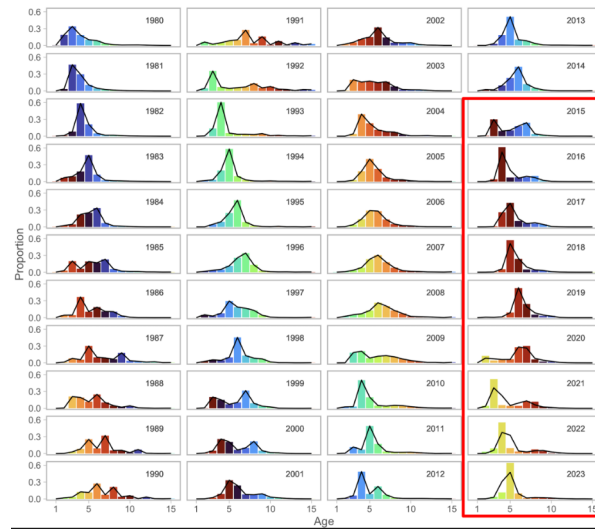


Figure 3: Proportions at age for recent years for the fishery data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

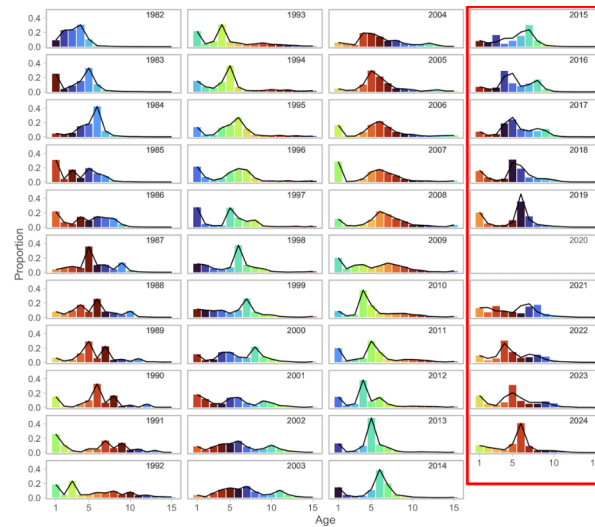


Figure 4: Proportions at age for recent years for the bottom-trawl survey data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

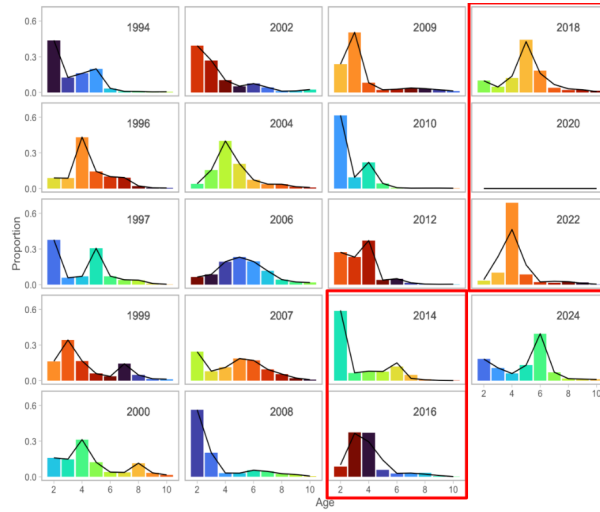


Figure 5: Proportions at age for recent years for the acoustic-trawl survey data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

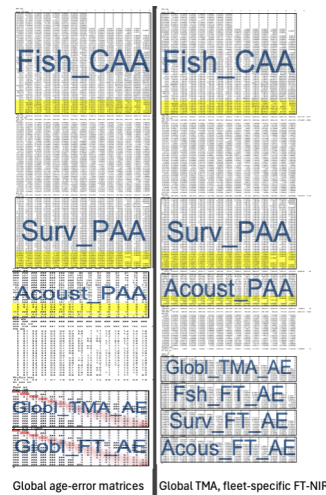


Figure 6: Schematic of input datafile showing the breakout of the fleet-specific FTNIRs age-error matrices (right panel) compared to that where the global FTNIRs age-error matrices are used (left panel).

Table 1: Configuration of different model options testing the impact of different treatment of the age composition data and ageing error.

Name	Model description	
	Age type	Age-error
Base	TMA	Global TMA
FT-NIR 1	TMA + FT-NIR	Global TMA, global FT-NIR
FT-NIR 2	TMA + FT-NIR	Global TMA, disaggregated FT-NIR

Table 2: Goodness-of-fit measures to primary data for different data and age-error applications. RMSE=root-mean square log errors, NLL=negative log-likelihood, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data). See text for incremental model descriptions.

Component	Base	FT-NIR fleets aggregated	FT-NIR fleet-specific
RMSE BTS	0.150	0.151	0.152
RMSE ATS	0.167	0.163	0.158
RMSE AVO	0.216	0.217	0.215
RMSE CPUE	0.097	0.097	0.097
SDNR BTS	0.890	0.900	0.880
SDNR ATS	0.930	0.910	0.880
SDNR AVO	0.950	0.960	0.940
Eff. N Fishery	358	336	381
Eff. N BTS	159	148	153
Eff. N ATS	233	221	233
Catch NLL	3	4	3
BTS NLL	24	24	23
ATS NLL	8	7	7
ATS Age1 NLL	7	8	9
AVO NLL	8	8	8
Fish Age NLL	417	559	614
BTS Age NLL	286	330	313
ATS Age NLL	43	42	41
NLL selectivity	213	216	215
NLL Priors	20	20	20
Data NLL	807	995	1,032
Total NLL	1110.900	1298.930	1332.810

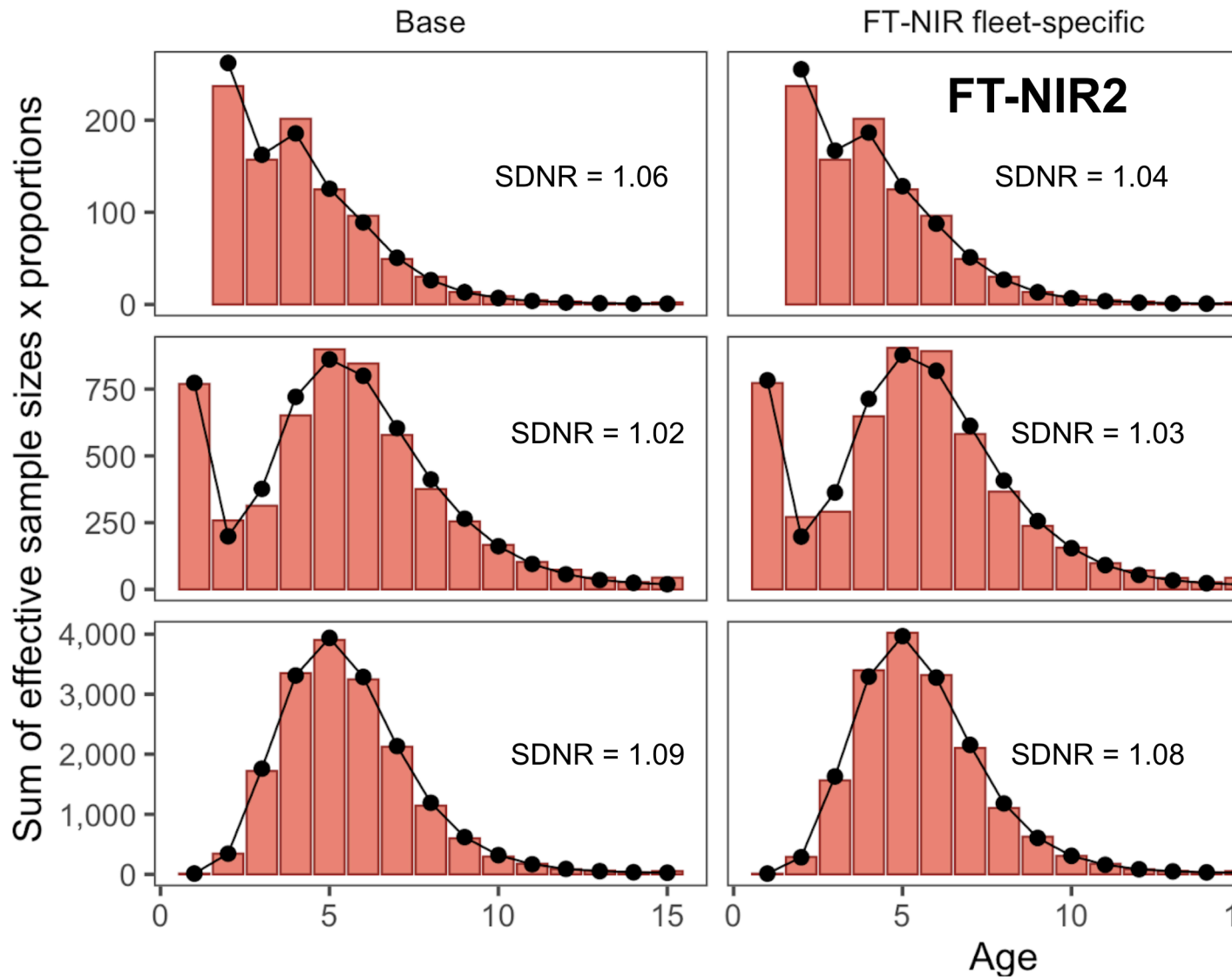


Figure 7: One-step ahead residuals for different model specifications for the fishery composition data

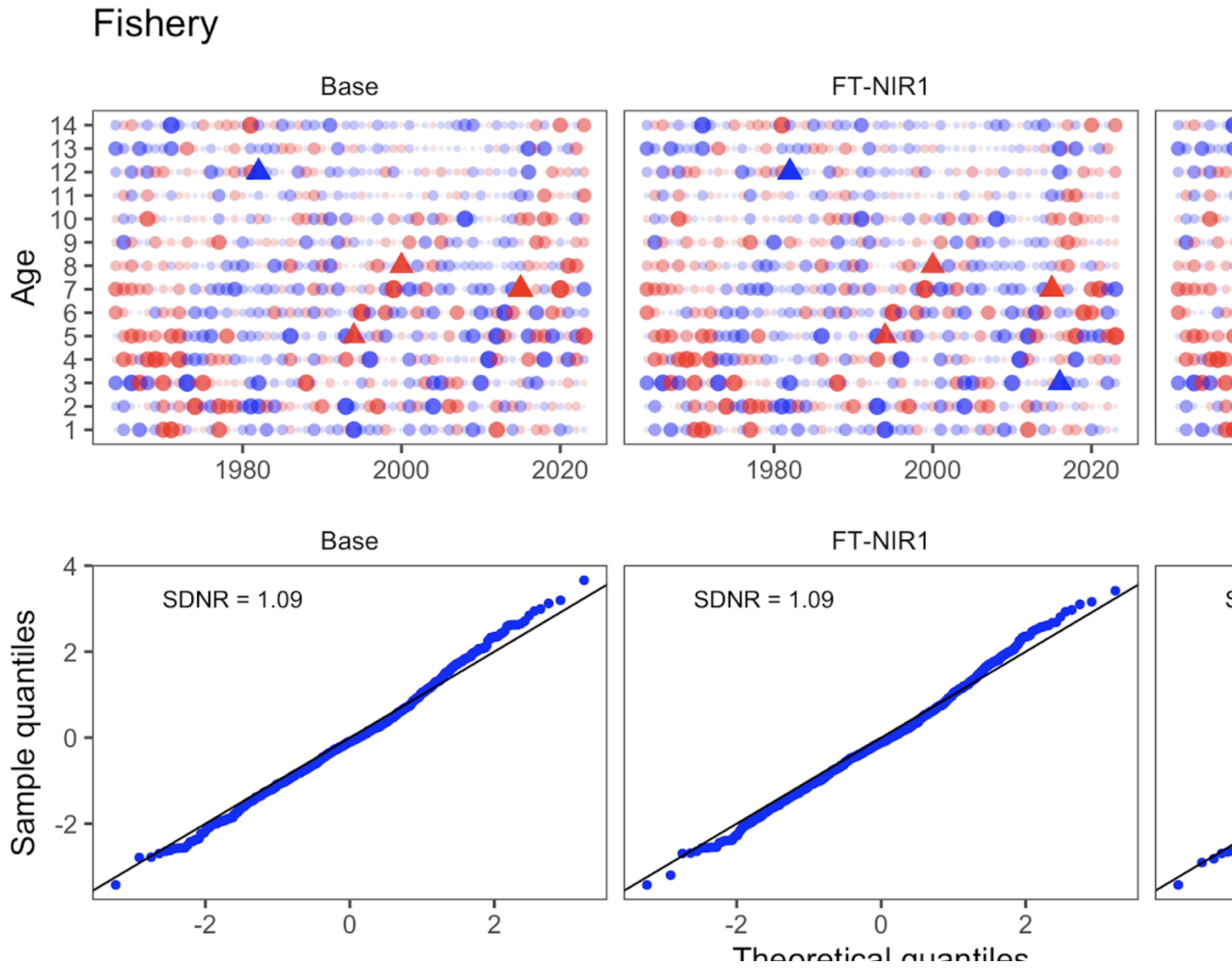


Figure 8: One-step ahead residuals for different model specifications for the fishery composition data

Bottom trawl survey

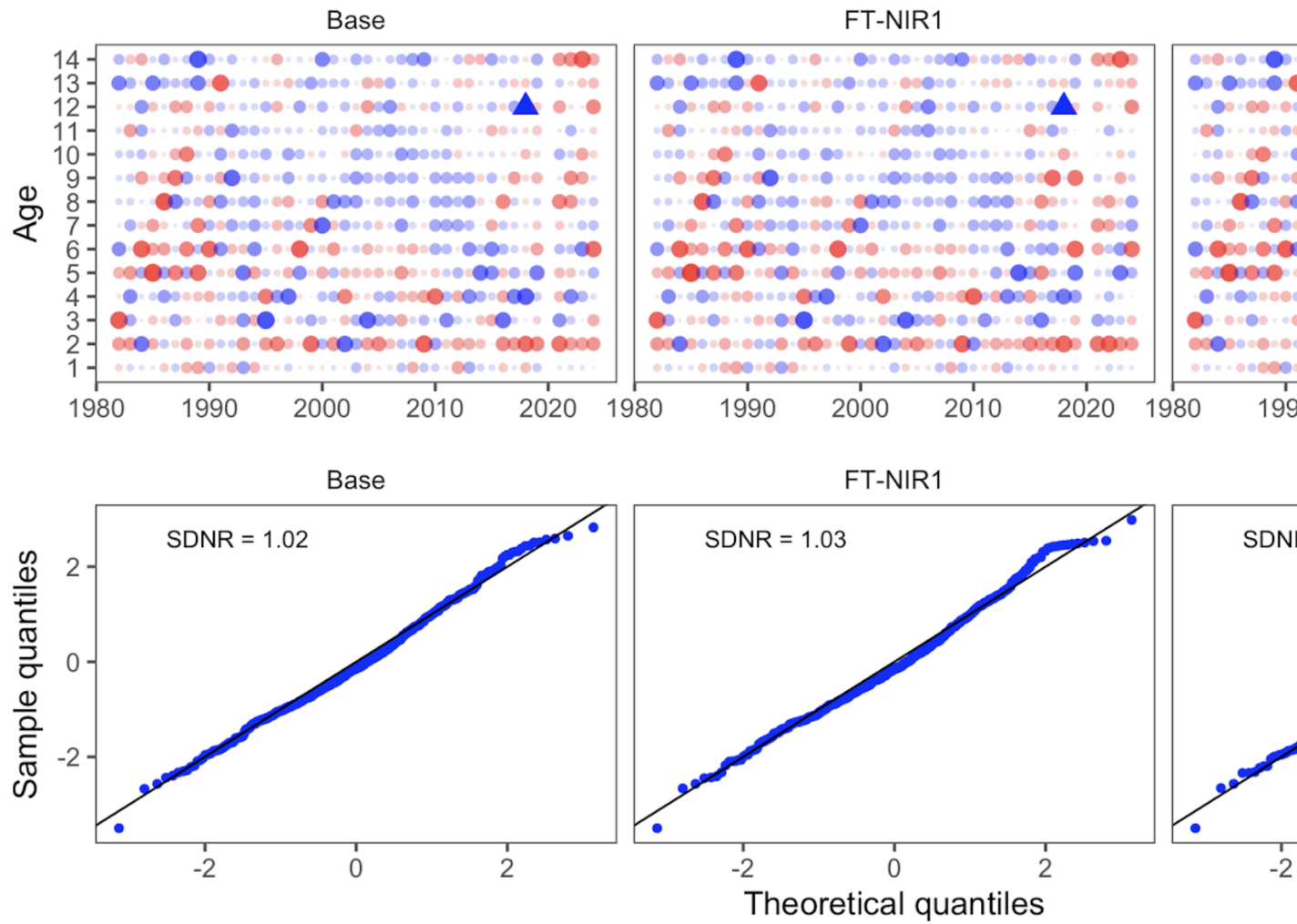


Figure 9: One-step ahead residuals for different model specifications for the bottom-trawl survey composition data

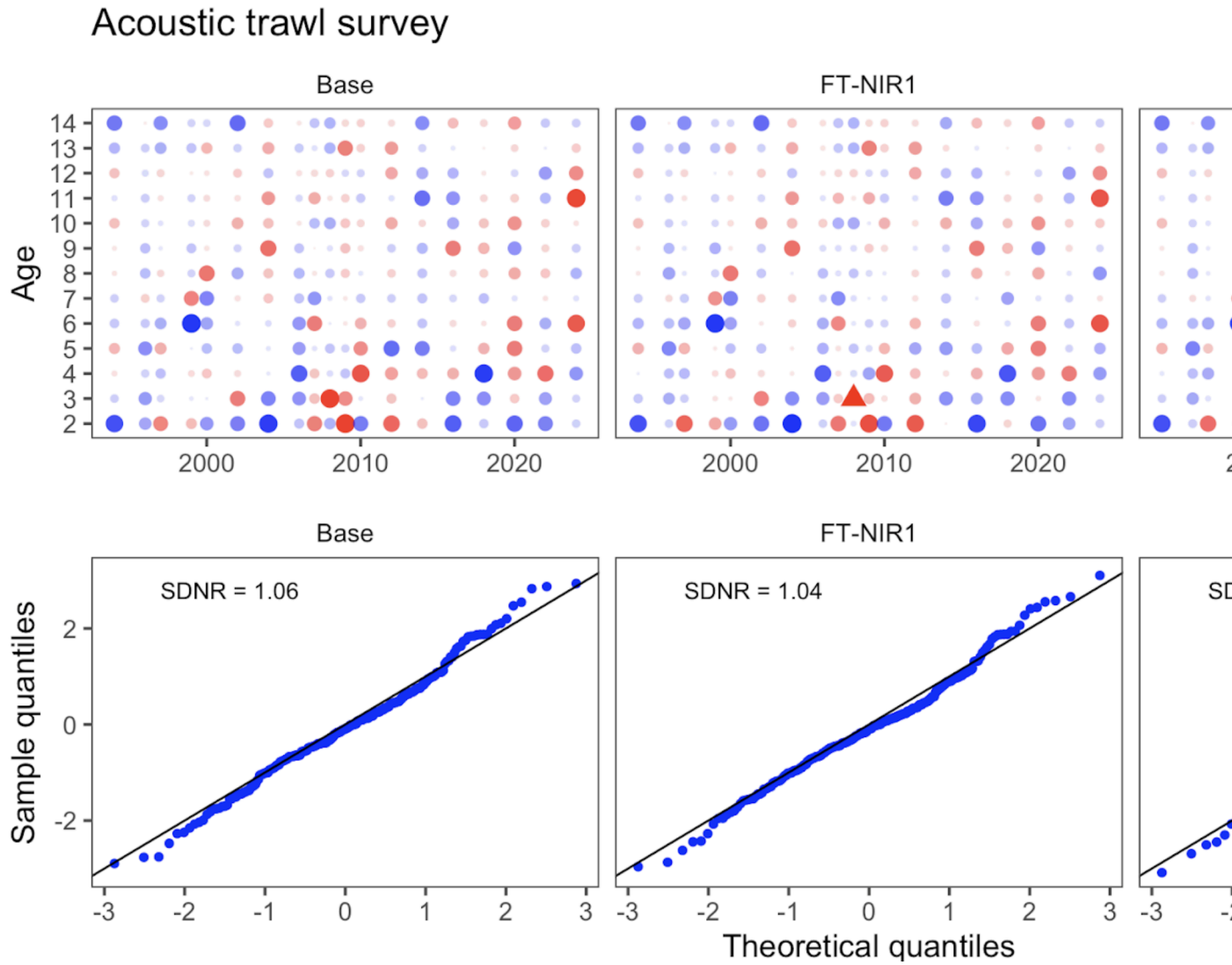


Figure 10: One-step ahead residuals for different model specifications for the acoustic-trawl survey composition data

Table 3: Goodness-of-fit measures to primary data for different data and age-error applications. RMSE=root-mean square log errors, NLL=negative log-likelihood, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data). See text for incremental model descriptions.

Component	Base	FT-NIR fleets aggregated	FT-NIR fleet-specific
B_{2025}	2,800	2,500	2,700
$CV_{B_{2025}}$	0.14	0.14	0.14
B_{MSY}	1,667	1,652	1,670
$CV_{B_{MSY}}$	0.23	0.24	0.23
B_{2025}/B_{MSY}	165%	153%	162%
B_0	4,318	4,360	4,436
$B_{35\%}$	1,935	1,898	1,931
SPR rate at F_{MSY}	31%	30%	29%
Steepness	0.63	0.64	0.64
Est. $B_{2024}/B_{2024, \text{nofishing}}$	0.65	0.62	0.64
B_{2024}/B_{MSY}	182%	169%	178%

Leave-one-out survey sensitivity analysis

Here we present model runs that downweight one or more sources of survey data at a time. Table 1 shows the combinations of data types included for this analysis, which includes six model runs: (1) downweight all Acoustic Trawl Survey (ATS) data (age 1 survey index, age 2+ survey biomass index, and age composition data), (2) downweight the ATS age composition data, (3) downweight the Acoustic Vessels of Opportunity (AVO) data (survey biomass index), (4) downweight all Eastern Bering Sea Trawl Survey (BTS) data (age 1+ survey biomass index and age composition data), (5) downweight the BTS age composition data, and (6) downweight all survey biomass indices (BTS, ATS, and AVO survey biomass indices). The ATS age 1 data are included as a separate survey index in the assessment model, while BTS age 1 data are included with the age composition data in the multinomial objective function component.

Among these models, downweighting the BTS data (both solely downweighting the BTS age composition data and downweighting all BTS data) led to changes in trends over time in estimated spawning biomass Figure 11 . The model estimates of recruitment were variable across sensitivity runs, but estimates differed the most from the 2024 accepted model when all BTS data were downweighted Figure 12. Estimates of BTS survey biomass fit the BTS data very poorly in most years when the BTS data were downweighted (Figure 13). Additionally, the fits to the BTS survey biomass were affected by downweighting the BTS age composition data (without downweighting the BTS survey index). The model was relatively insensitive to other data being excluded, and fits to the ATS and AVO indices were relatively insensitive to downweighting the BTS data (see Figure 13, Figure 14, and Figure 15). The root mean square error (RMSE) in Table 2 for the model run that downweighted all BTS data was large and the effective sample size (Eff. N) was low as compared to the 2024 accepted model.

Table 1: Configuration of different model options testing the impact of different data components

Dataset	Model description					
	Drop all BTS	Drop all ATS	Drop all indices	Drop AVO	Drop BTS Age composition	Drop ATS Age composition
BTS age 2+ Index	X		X			
ATS age 2+ Index		X	X			
ATS age 1 Index		X	X			
AVO Index			X			
BTS comps	X				X	
ATS comps		X				

Additionally, downweighting solely the BTS age composition data led to low effective sample size for the BTS.

The importance of the survey data can be seen in some of the reference calculations where the uncertainty increases dramatically and the stock-status perception was quite different across sensitivity runs Table 3. Most dramatically, dropping all survey indices led to a CV in “ B_{MSY} ” estimates of 1.26 as compared to 0.13 in the 2024 accepted model.

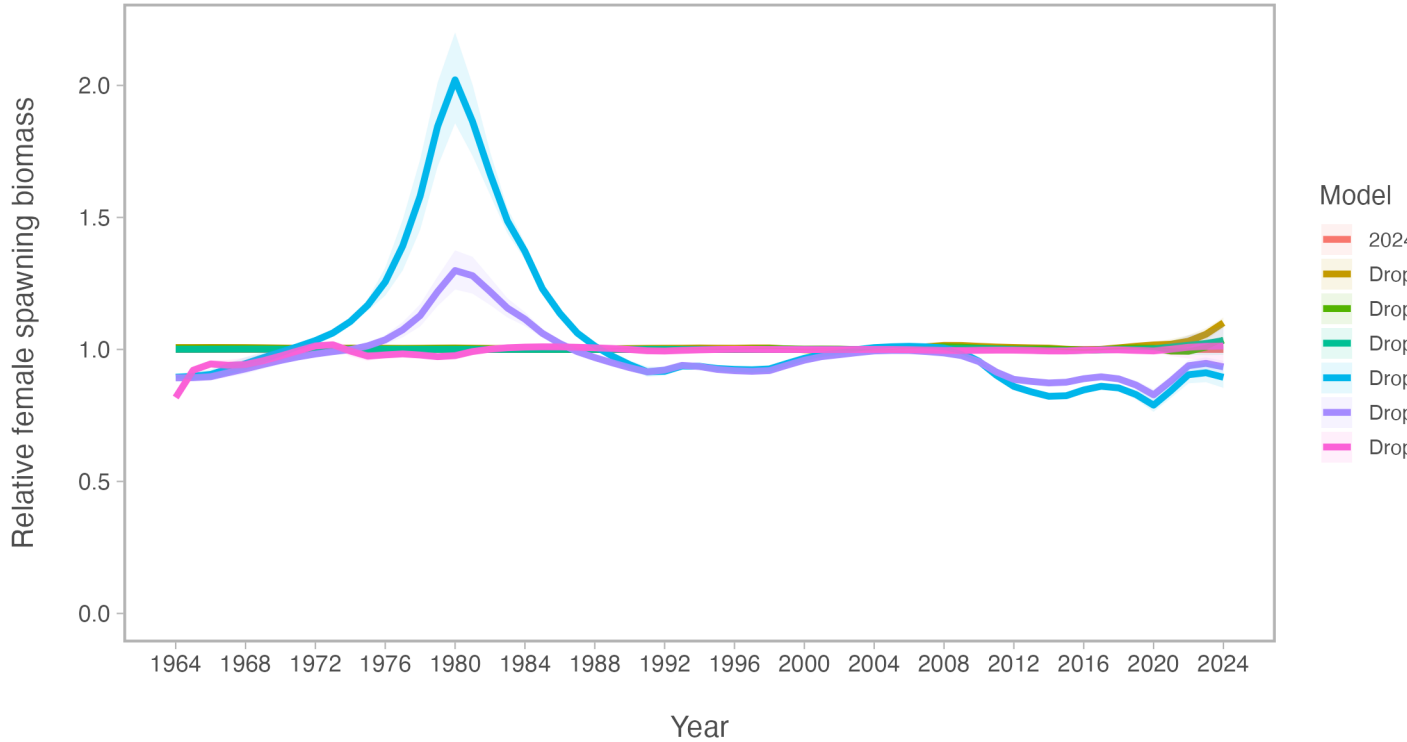


Figure 11: Comparison of spawning biomass (SSB) relative to the 2024 accepted model for models that downweight one data source at a time

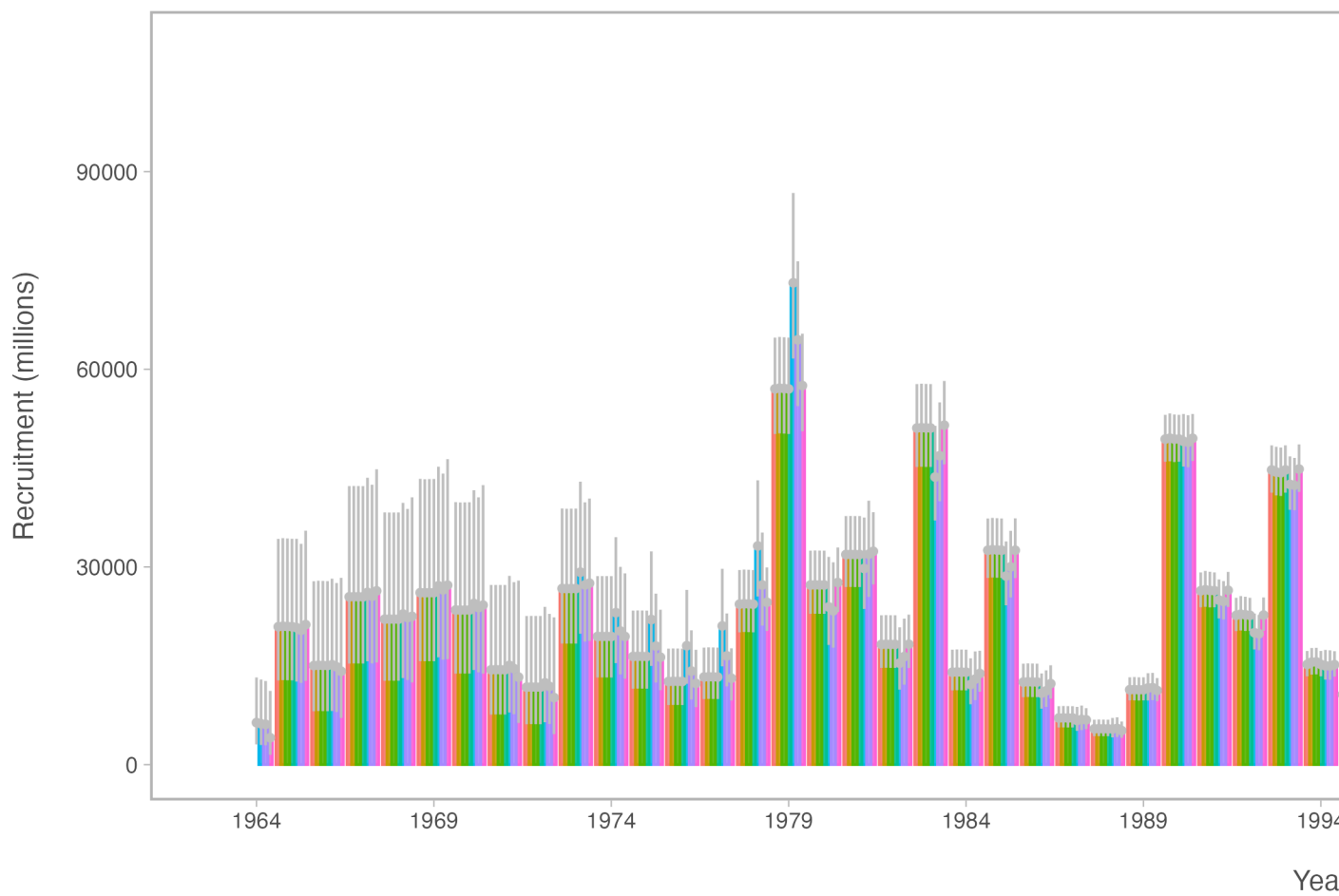


Figure 12: Comparison of recruitment estimates for models that downweight one data source at a time

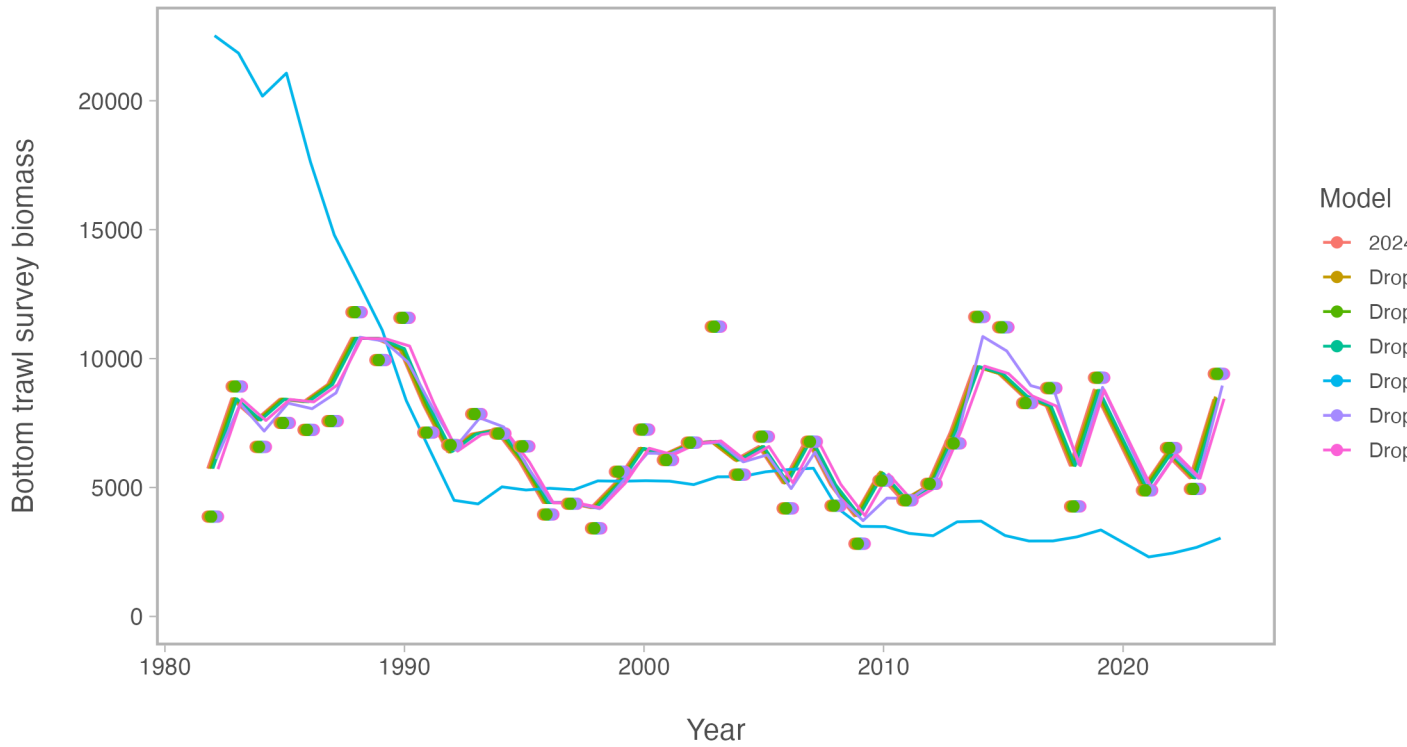


Figure 13: Comparison of fits to the Eastern Bering Sea Bottom Trawl Survey (BTS) biomass index for models that downweight one data source at a time

Table 2: Goodness-of-fit measures to primary data for leave-one-out model sensitivity runs. RMSE=root-mean square log errors, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data).

Component	2024 accepted	Drop BTS	Drop ATS	Drop AVO	Drop indices	Drop BTS age
RMSE BTS	0.157	0.689	0.157	0.157	0.157	0.152
RMSE ATS	0.180	0.182	0.188	0.191	0.187	0.178
RMSE AVO	0.229	0.237	0.238	0.241	0.238	0.231
RMSE CPUE	0.093	0.090	0.093	0.093	0.098	0.092
Eff. N Fishery	1,182	1,379	1,190	1,185	1,176	1,344
Eff. N BTS	248	95	254	249	245	84
Eff. N ATS	269	230	164	269	273	228

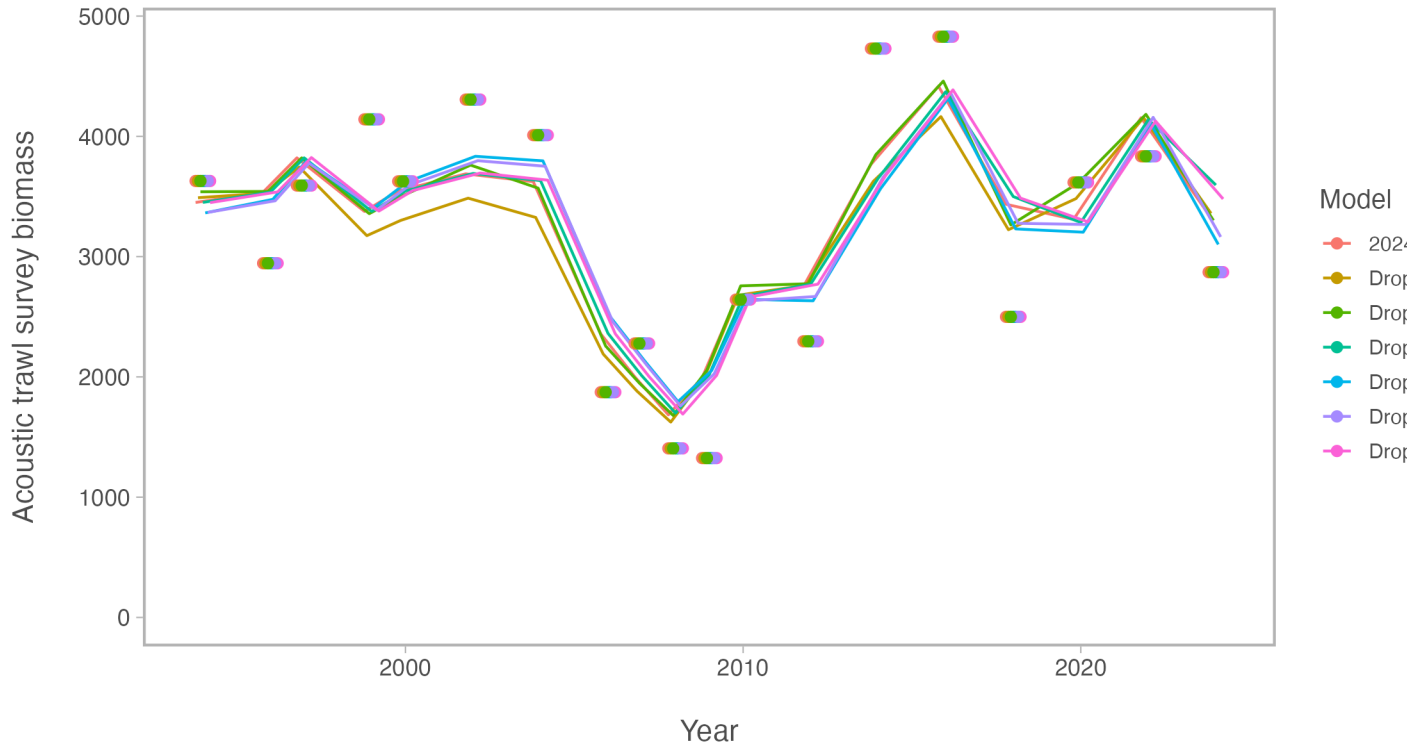


Figure 14: Comparison of fits to the Acoustic Trawl Survey (ATS) biomass index for models that downweight one data source at a time

Table 3: Estimated derived quantities and selected corresponding CVs for leave-one-out sensitivity analysis runs. See text for incremental model descriptions.

Estimated Quantity	2024 accepted	Drop BTS	Drop ATS	Drop AVO	Drop indices	Drop
B_{2025}	3,100	2,700	3,600	3,300	3,200	
$CV_{B_{2025}}$	0.13	0.16	0.14	0.16	0.18	
B_{MSY}	2,310	2,415	2,360	2,322	3,677	
$CV_{B_{MSY}}$	0.3	0.31	0.31	0.31	1.26	
B_{2025}/B_{MSY}	135%	111%	151%	141%	86%	
B_0	5,975	6,136	6,120	6,009	8,934	
$B_{35\%}$	2,066	1,988	2,101	2,080	2,074	
SPR rate at F_{MSY}	31%	35%	31%	31%	42%	
Steepness	0.62	0.59	0.62	0.62	0.52	
Est. $B_{2024}/B_{2024, \text{nofishing}}$	0.61	0.51	0.63	0.61	0.42	
B_{2024}/B_{MSY}	147%	126%	158%	151%	93%	

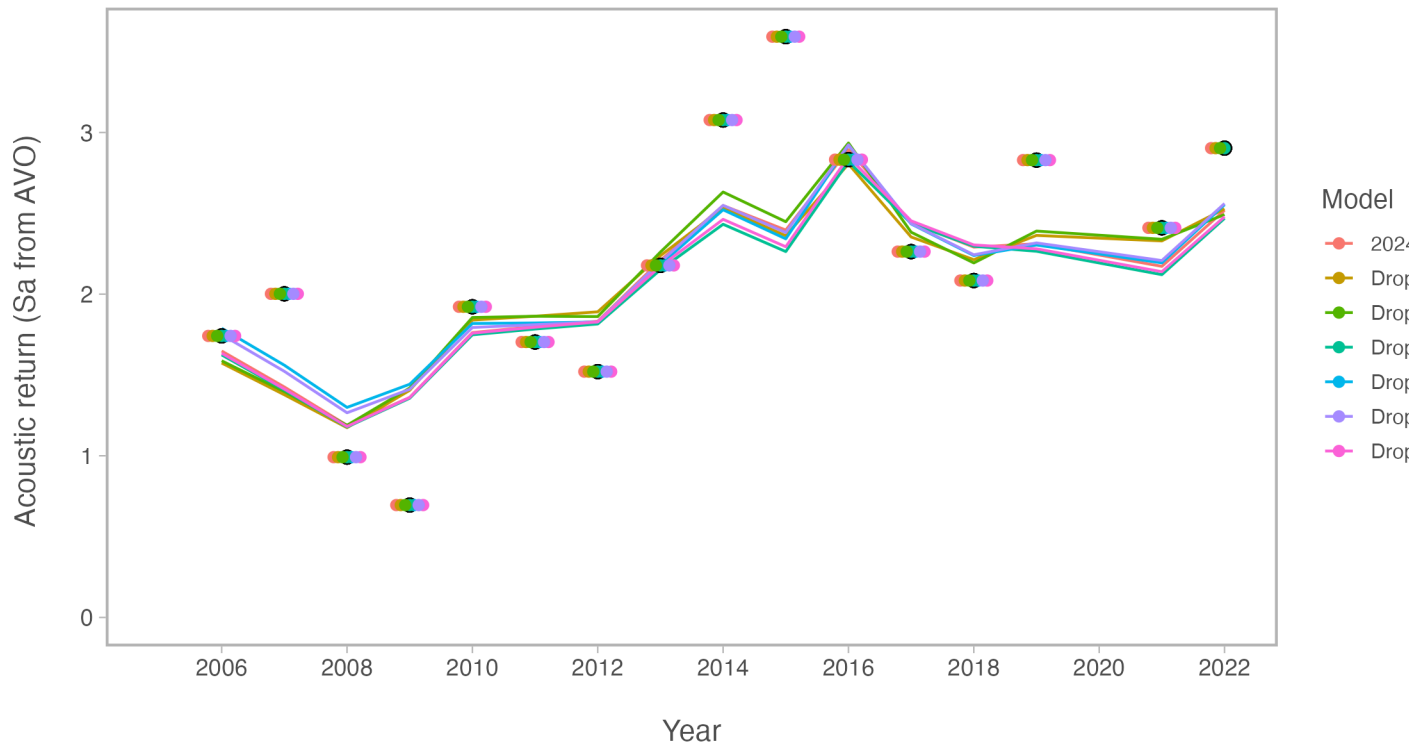


Figure 15: Comparison of fits to the Acoustic Vessels of Opportunity (AVO) biomass index for models that downweight one data source at a time

Alternative assessment software

Stock-synthesis 3

Stock synthesis 3 (SS3) is a well establish framework and is common in many diverse settings. (Methot and Wetzel (2013)) For EBS pollock, we extended an initial framework based on the configurations used for Pacific hake (Grandin et al. (2024)) since there are many similarities in the type of data available. Namely, empirical weight-at-age, acoustic survey data, and indices of 1-year olds.

Within the SS3 framework, we evaluated different configurations for fishery selectivity variability (Figure 16). These showed the impact on the uncertainty and stock trends (Figure 17 for SSB and Figure 18 for recruitment).

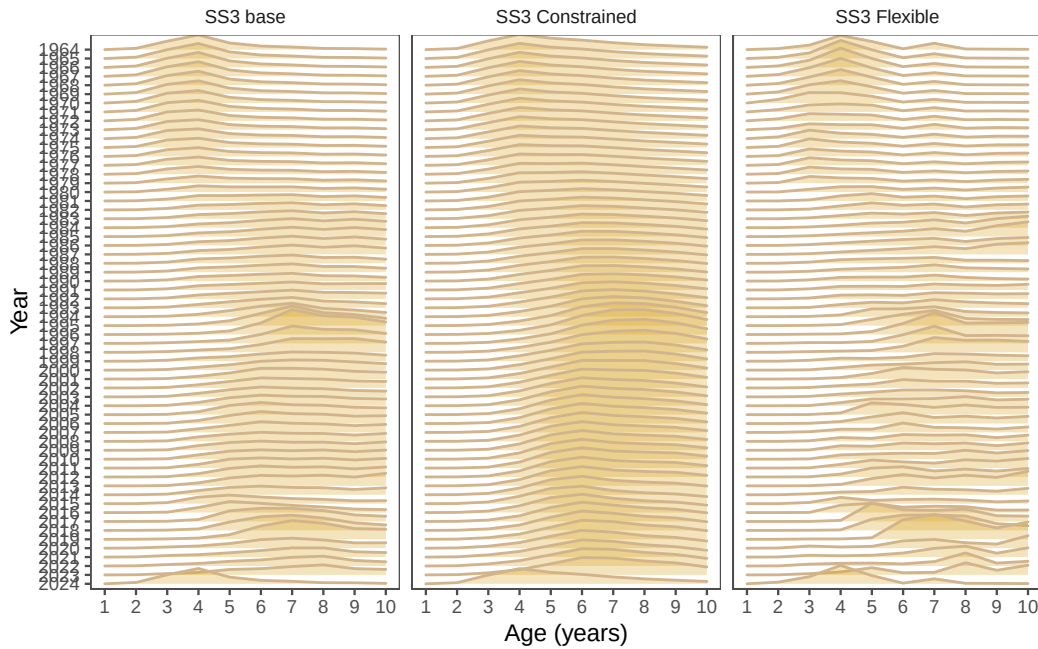


Figure 16: Model results comparing three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

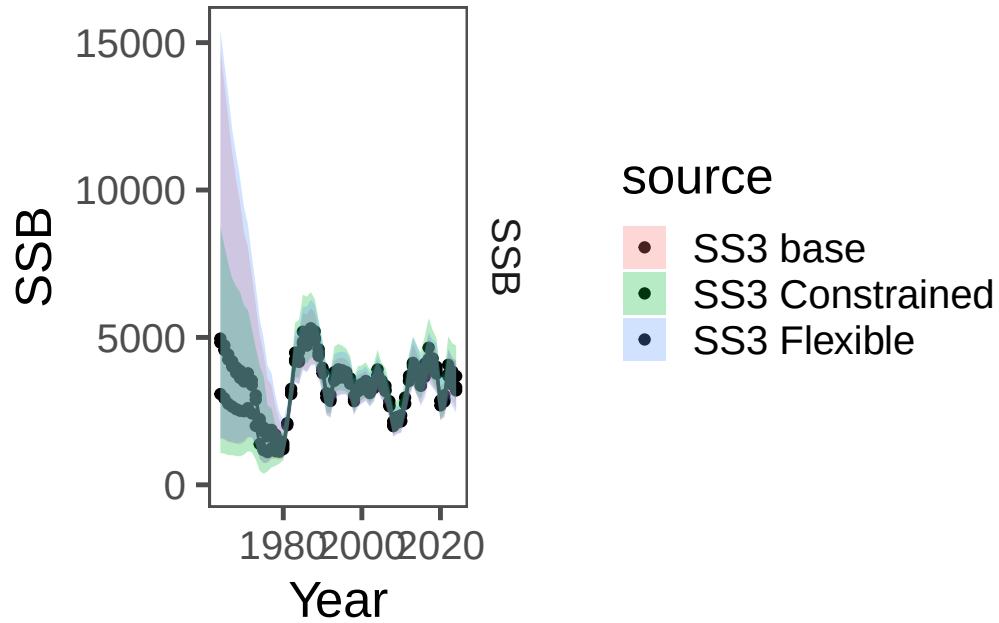


Figure 17: Model results comparing the time-series for three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

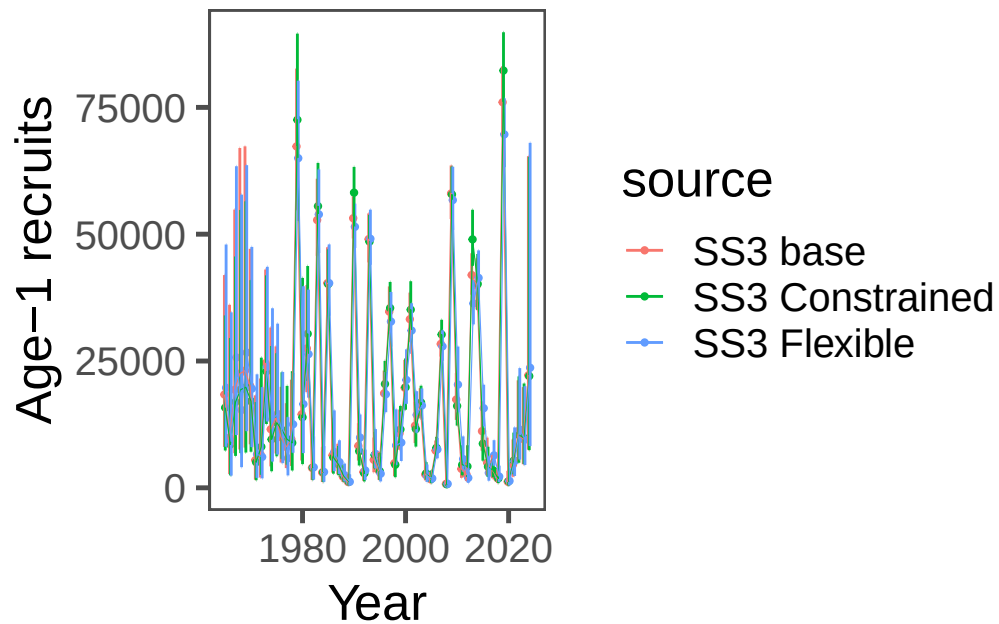


Figure 18: Model results comparing the time-series for three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

Comparing the “base” SS3 model with the pollock model showed different patterns in the early periods for selectivity (Figure 19) but slight differences in SSB and recruitment estimates (Figure 20 and Figure 21).

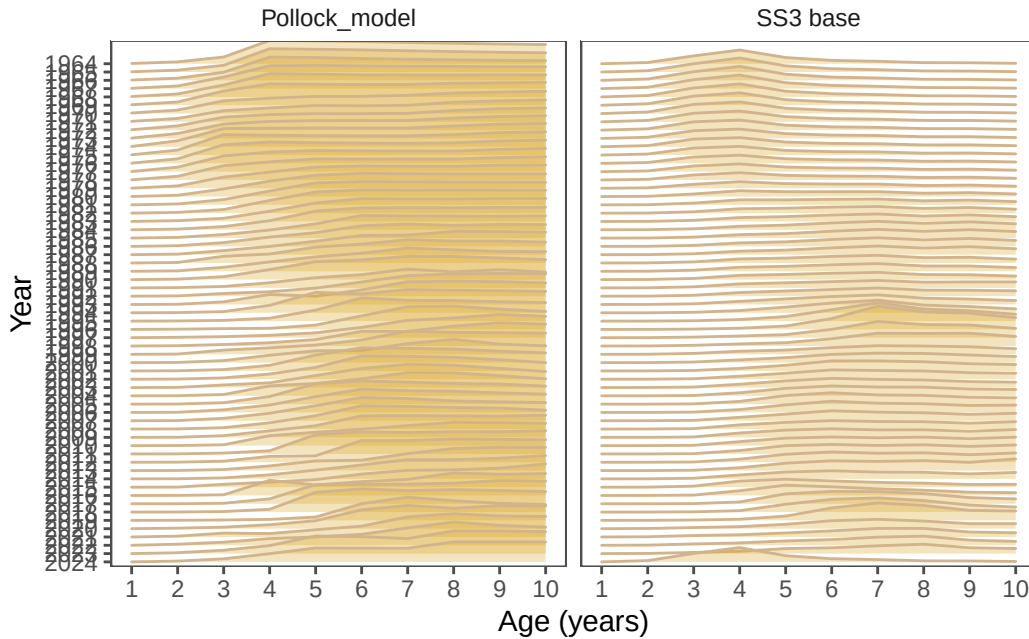


Figure 19: Model results comparing selectivity estimates between the base SS3 model configuration and the pollock model accepted in 2024.

SAM

The Stock Assessment Model (SAM) is a state-space model written in TMB (Nielsen and Berg (2014), Nielsen et al. (2021)). We attempted to configure the settings to have similar levels of flexibility to the reference model used for catch advice in 2024 and the settings and results can be found [here](#)). Comparing “selectivity” in the SAM model to the reference model, we see broad similarities but slightly lower inter-annual variability in the SAM run (Figure 22).

The spawning biomass trend largely overlaps in the uncertainty between the SAM and pollock model (Figure 23). The recruitment estimates are similar as well (Figure 24).

Other models in development

The “Woods Hole Assessment model” (WHAM) is written in TMB and is a fully state-space model (Stock and Miller (2021)). Initial trials are available on the [github repository](#).

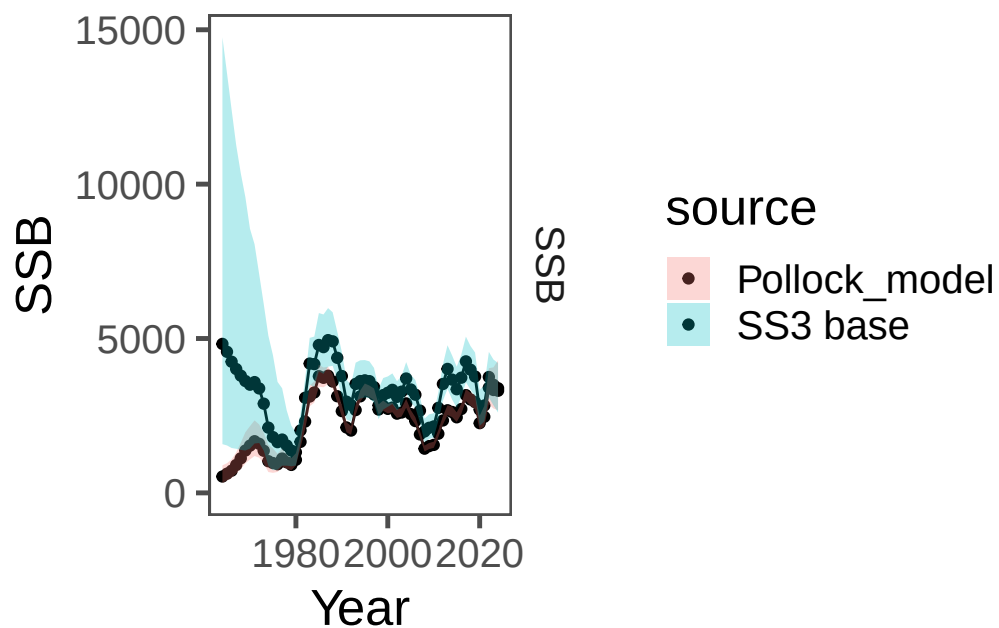


Figure 20: Model results comparing the base SS3 model configuration with the model used for pollock in 2024.

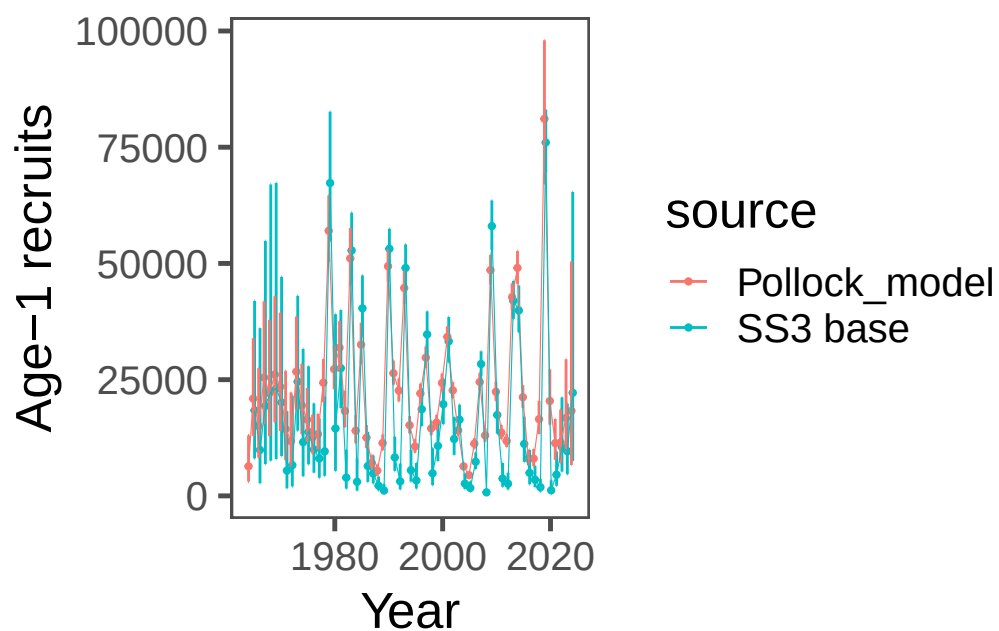


Figure 21: Model results comparing the base SS3 model configuration with the model used for pollock in 2024.

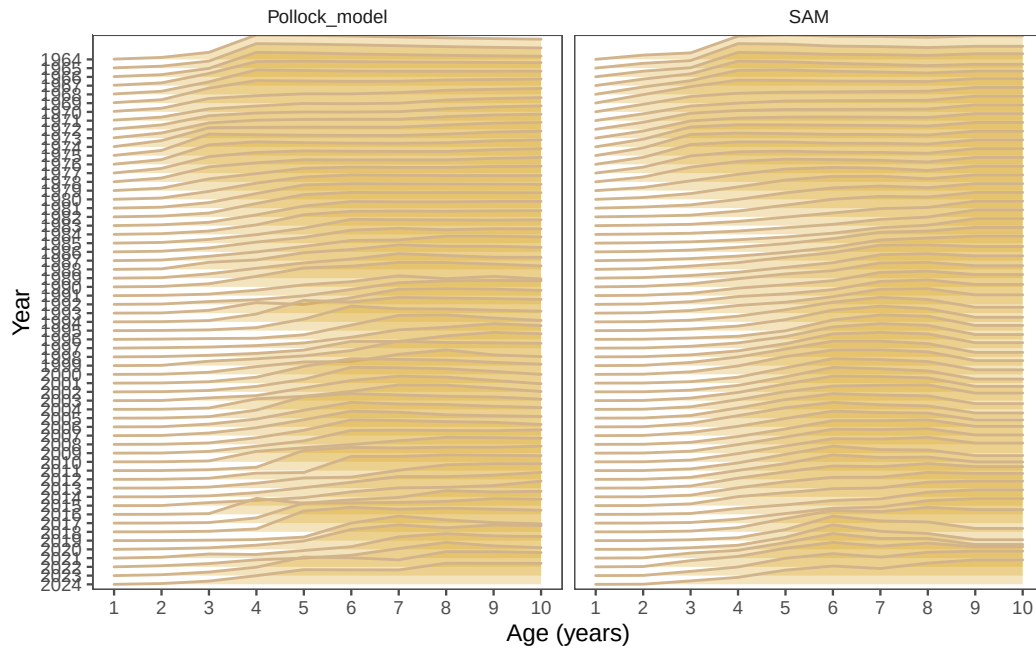


Figure 22: Model results comparing last year's model with the SAM model.

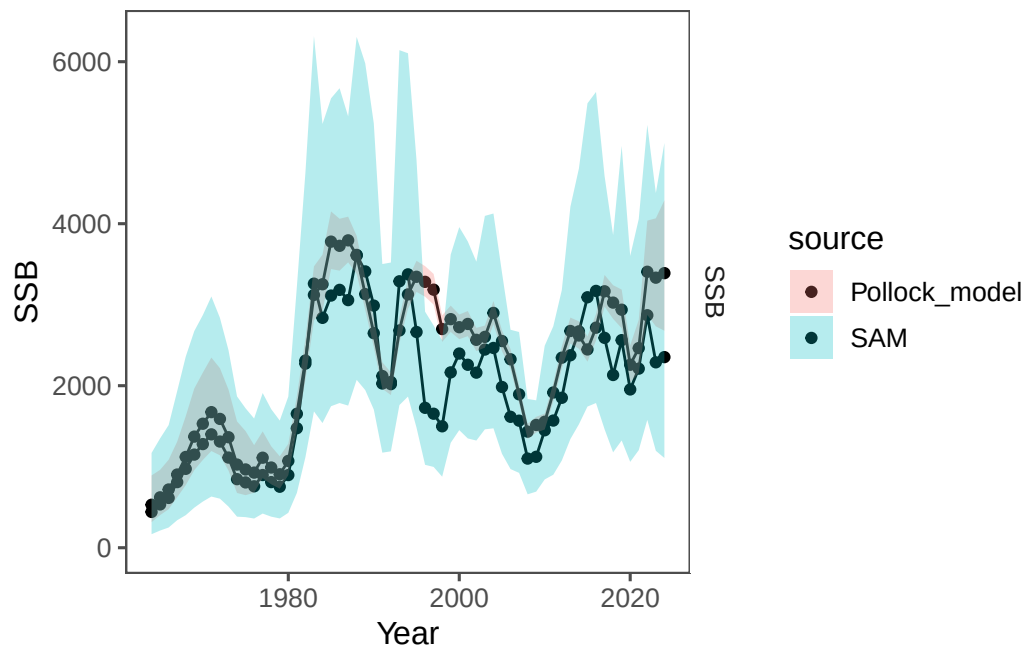


Figure 23: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

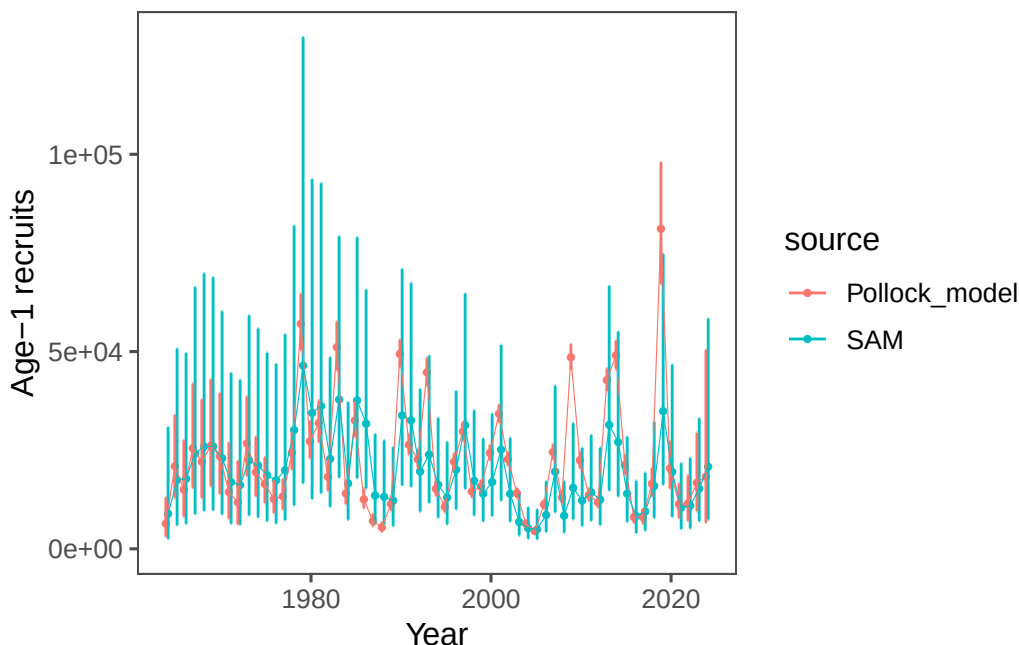


Figure 24: Model results comparing last year’s model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

The Rceattle model is developed by the AFSC and is written in TMB (Adams et al. (2022)). The model is an extension of the ADMB version used for multi-species model in the Eastern Bering Sea (Jurado-Molina, Livingston, and Ianelli (2005), Holsman et al. (2024)) and has a number of newer features. It is generalizable to include a variety of data types using lengths and can be set up for species tracked for sexual dimorphic growth and other processes. Since it has been recast into TMB, it allows for the ability to specify random effects in a number of processes. One development of the model is to have the flexibility to bridge among extant single-species modeling platforms by using it in “single-species” mode. To illustrate that, we introduce an application of Rceattle to the EBS pollock data set. The model is not yet fully developed, but we present preliminary results for consideration. Initial runs are favorable but further work is needed to bridge the selectivity options (Figure 25).

Spatial modeling of the EBS pollock stock

In development is a spatially disaggregated general model by colleagues at the Auke Bay lab and the University of Alaska, Fairbanks (M. Cheng (2025)). This general model which has flexibility to specify regions and a variety of options including random effects. While this model is not yet fully developed, we present preliminary results here. The model is written in RTMB and is similar to the WHAM model. We pursue this to explore the potential for a more

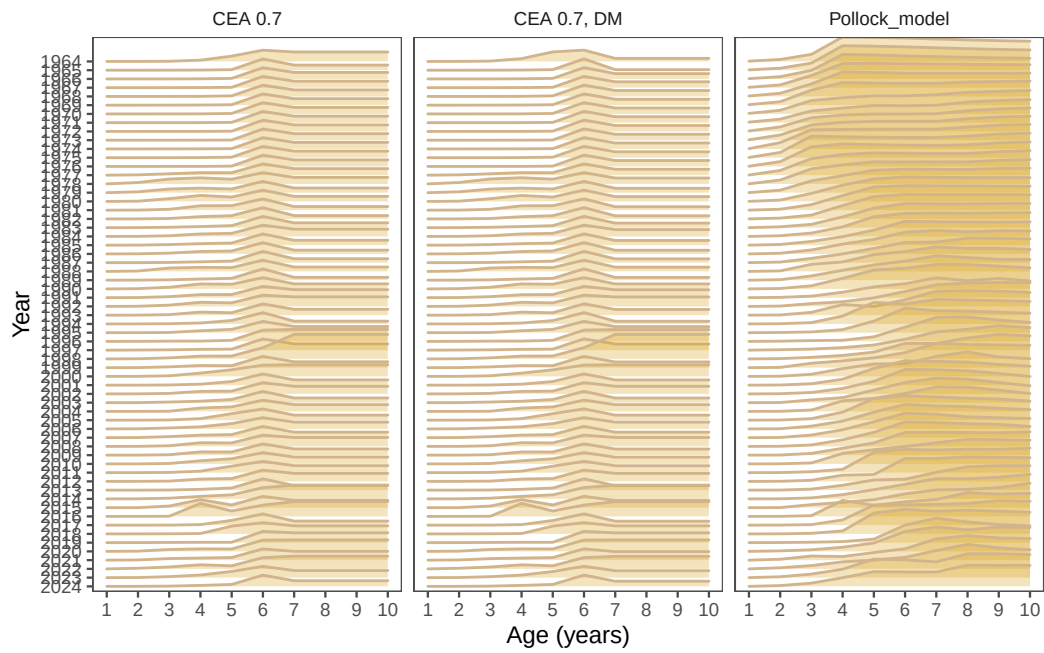


Figure 25: Model results comparing last year's model with the Rceattle model.

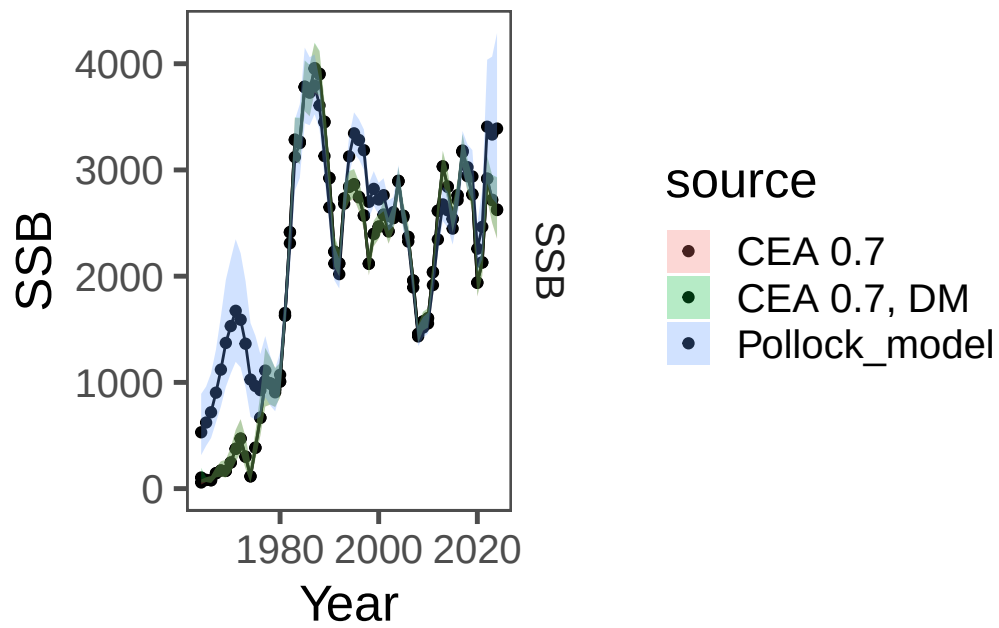


Figure 26: Model results comparing last year's model with the Rceattle model.

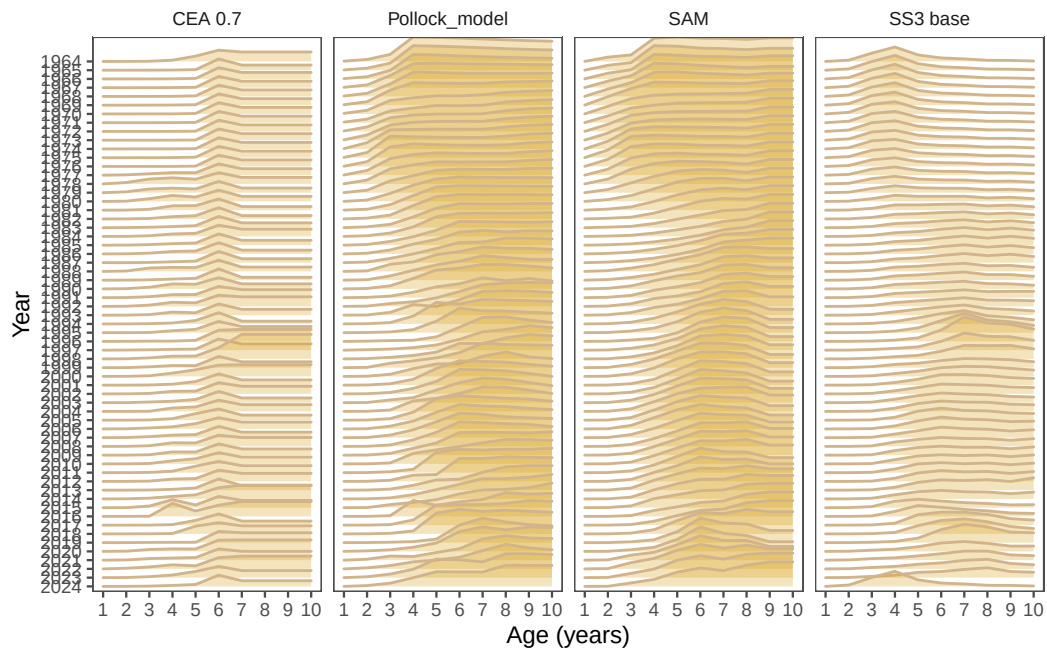


Figure 27: Model results comparing last year's model with the Rceattle model.

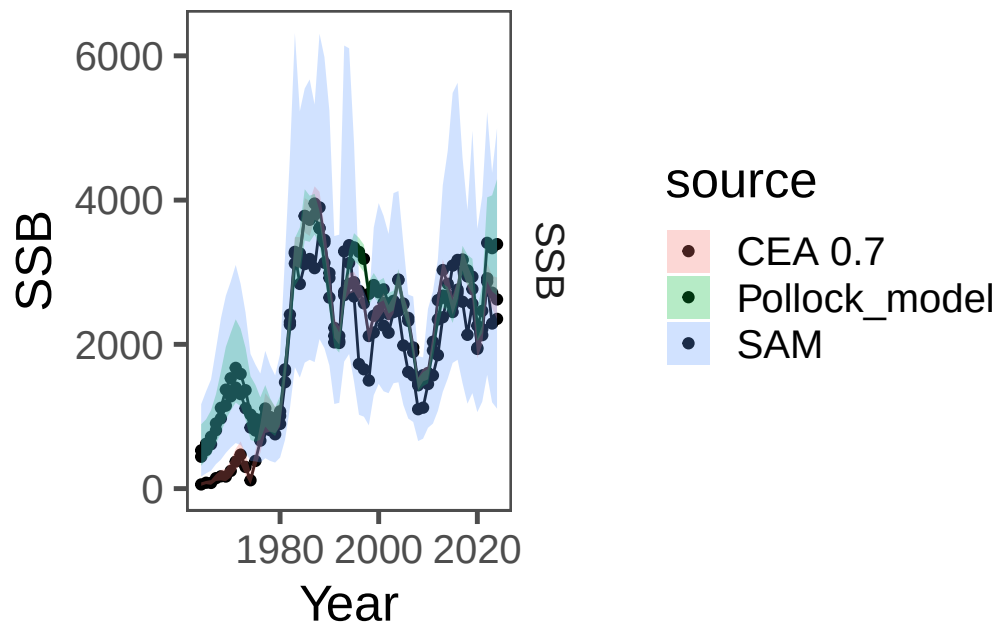


Figure 28: Model results comparing last year's model with the Rceattle model.

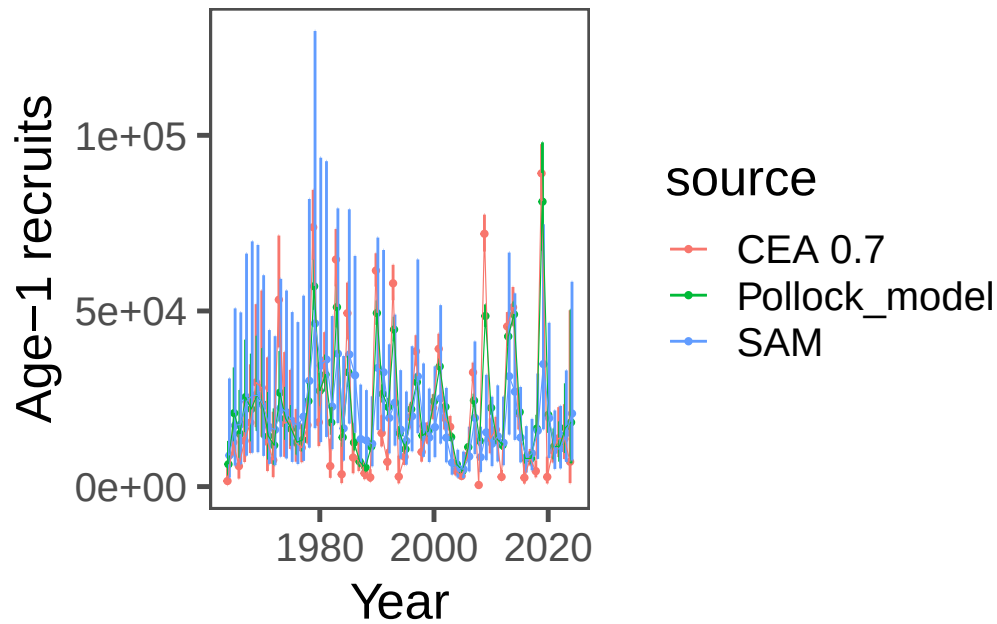


Figure 29: Model results comparing last year's model with the Rceattle model.

flexible model that can be used for more explicit considerations of stocks linked to different regions (e.g., the US and Russian portions of the EBS). Also, this model to add sex-specificity and random effects (M. L. H. Cheng et al. (2025)).

Figure 30 shows the model results comparing time-varying selectivity whereas Figure 31 shows the model results time series. The difference in the age-1 recruits (Figure 32) is likely due to the fact that the spatial model currently has constant M-at-age compared to the base model.

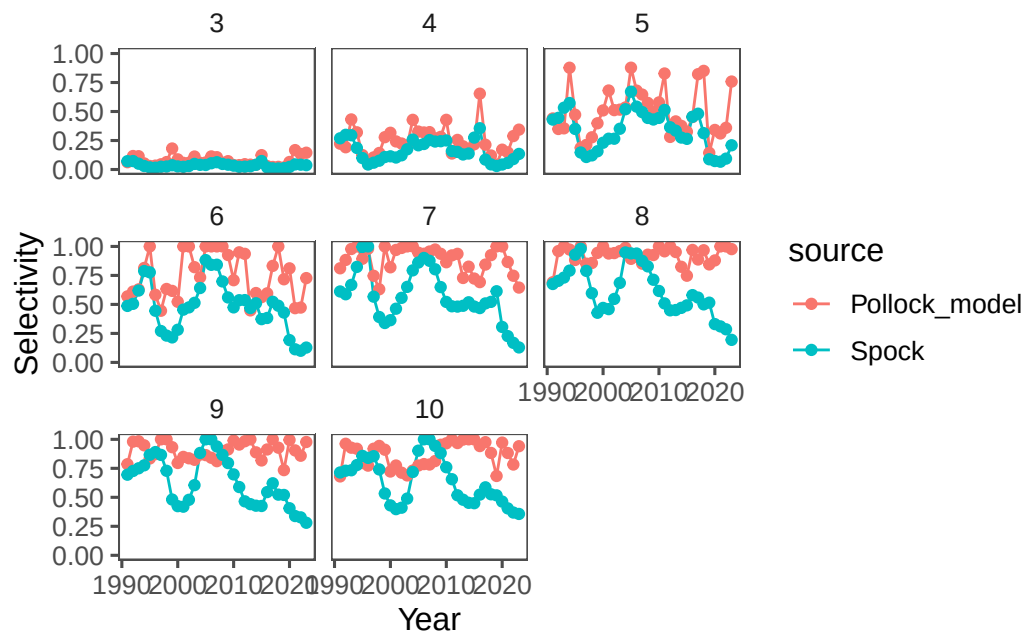


Figure 30: Model results comparing last year's model with the SPock model.

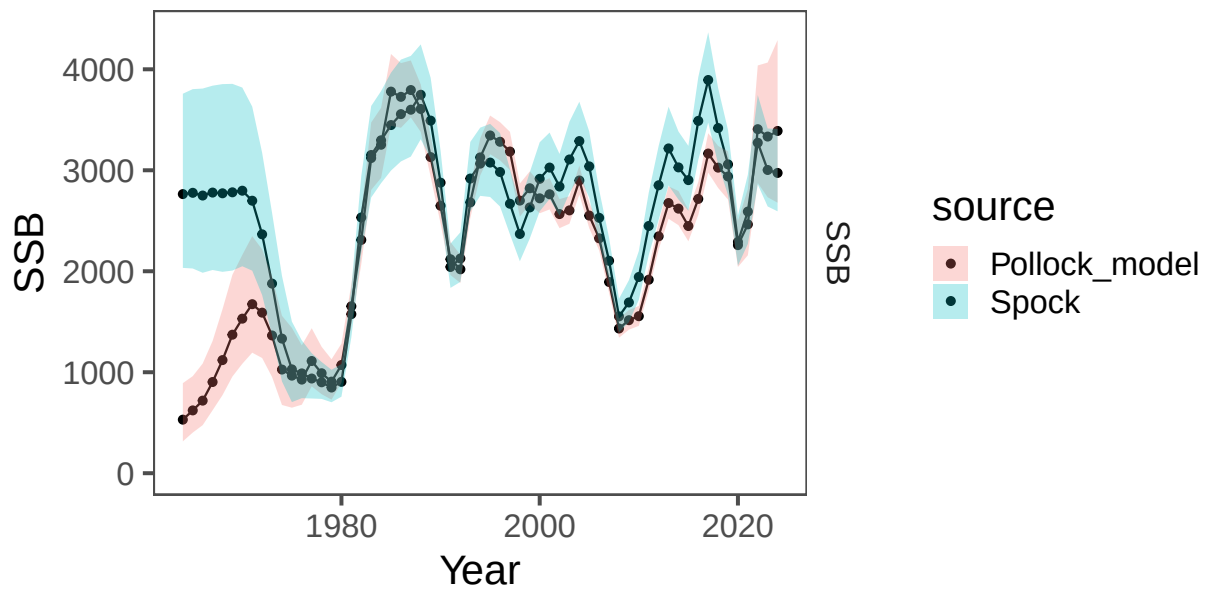


Figure 31: Model results comparing last year's model with the SPock model.

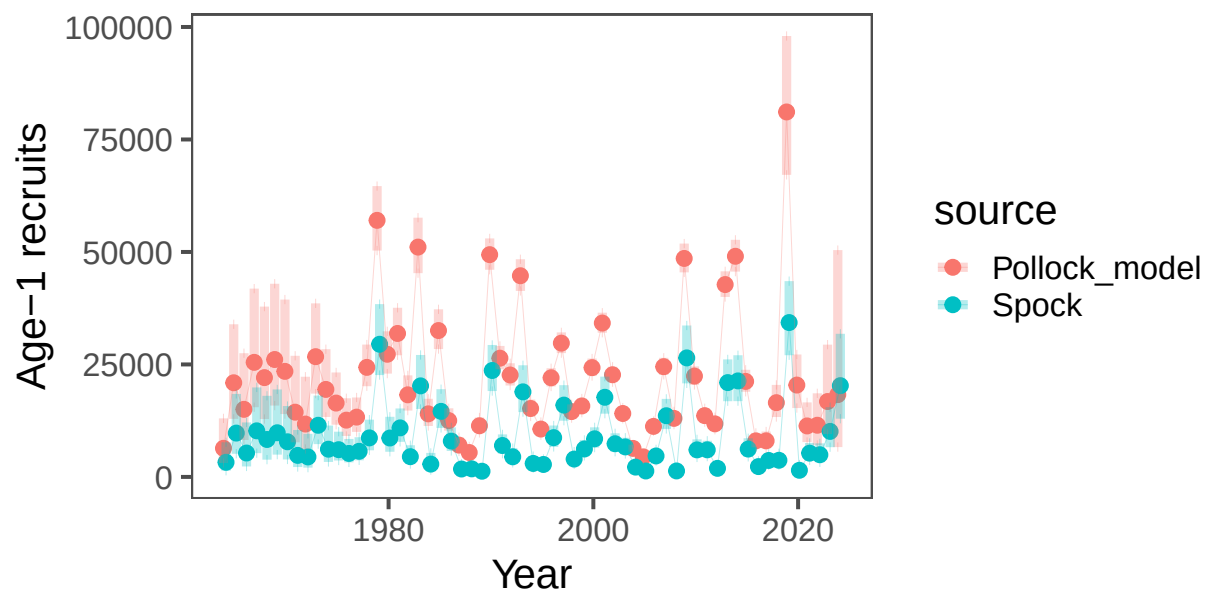


Figure 32: Model results comparing last year's model with the SPock model.

Bottom-trawl survey spatial distribution modeling

Abundance Index Methods

For model-based indices in the Bering Sea, we fitted observations of numerical abundance or biomass per unit area (where the choice of abundance or biomass varied by stock at the request of assessors) from all sampling locations in the 83-112 bottom trawl survey of the EBS, 1982-2025, including:

- Exploratory northern extension samples in 2001, 2005, and 2006
- 83-112 samples available in the NBS in 1982, 1985, 1988, 1991, 2010, 2017-2019, 2021-2023, and 2025

NBS samples collected prior to 2010 and in 2018 did not follow the 20 nautical mile sampling grid that was used in other survey years. Assimilating these unbalanced data requires extrapolating into unsampled areas, facilitated by including a spatially varying response to cold-pool extent (Thorson 2019). This spatially varying coefficient term was estimated for both linear predictors of a delta-model, and detailed comparison of results for EBS pollock has shown that it has a small but notable effect on these indices and resulting stock assessment outputs (O’Leary et al. 2020).

Rather than using the cold-pool covariate for yellowfin sole, we instead used the mean bottom temperature within the inner and middle domain strata from an interpolated temperature product. All models were fitted in the sdmTMB R package (<https://pbs-assess.github.io/sdmTMB>; Anderson et al. (2022)). All environmental data used as covariates were computed within the coldpool R package (<https://github.com/afsc-gap-products/coldpool>; Rohan, Barnett, and Charriere (2022)).

We used a spatiotemporal Poisson-link delta-model (Thorson 2018) involving two linear predictors and a gamma distribution to model positive catch rates. We predicted population density across the entire EBS and NBS in each year, using AFSC GAP-approved prediction grids. Spatial and spatiotemporal fields were estimated using a spatial mesh including 250 “knots” whose locations were determined using a K-means algorithm to place the knots evenly over space, in proportion to the dimensions of the prediction grid.

While the prior VAST model was fitted with a 750 knot mesh, we found that sdmTMB models fitted with mesh resolutions from 250-750 knots all provided extremely similar index estimates and that reducing the number of knots dramatically improved computation time. We estimated geometric anisotropy (how spatial autocorrelation declines with differing rates over distance in some cardinal directions than others) and included a spatial and spatiotemporal term for both linear predictors.

To facilitate prediction of density in regions with unsampled years, we specified that the spatiotemporal fields were structured over time as an AR(1) process (where the magnitude of autocorrelation was estimated as a fixed effect for each linear predictor). However, in cases where the AR(1) correlation parameter was estimated to be close to 1 for either model component, the model would be collapsed into a simpler structure by specifying $\rho = 1$, i.e., modeling spatiotemporal variation as a random walk. We did not include any temporal correlation for intercepts, which we treated as fixed effects for each linear predictor and year. Finally, we used epsilon bias-correction to correct for retransformation bias (Thorson and Kristensen 2016).

We checked model fits for convergence by confirming that the derivative of the marginal likelihood with respect to each fixed effect was sufficiently small (less than ~ 0.001) and that the Hessian matrix was positive definite, along with other model checks produced by the function `sdmTMB::sanity()`. We then checked goodness of model fit by computing Dunn-Smyth randomized quantile residuals (Dunn and Smyth 1996) and visualizing these using a quantile-quantile plot within the DHARMa R package (Hartig 2021). We also evaluated the distribution of these residuals over space in each year, and inspected them for evidence of residual spatiotemporal patterns. Results compared favorably among methods, and the model index result was very similar to previous estimates (Figure 33).

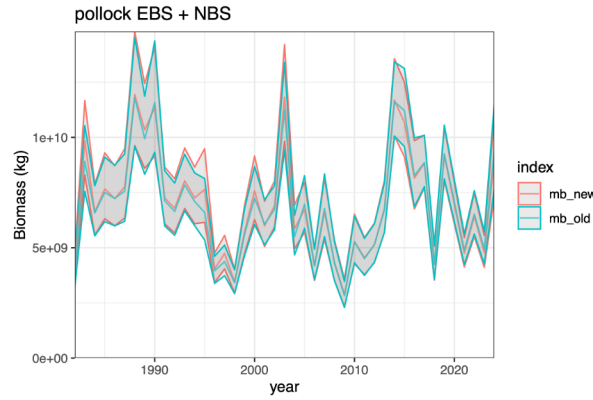


Figure 33: Model results comparing last year’s model with the revised sdmTMB model.

Model-based Age Composition Methods

For model-based estimation of age compositions in the Bering Sea, we fitted observations of numerical abundance-at-age at each sampling location. This was made possible by applying a year-specific, region-specific (EBS and NBS) age-length key to records of numerical abundance and length composition. In subcategories (combinations of year, length, age, sex) that contained insufficient data, age composition was computed from length composition given a globally pooled age-length key.

We computed these estimates in the `tinyVAST` R package (<https://vast-lib.github.io/tinyVAST>; Thorson et al. (2024)), assuming a Poisson-link delta-model (Thorson 2018) involving two linear predictors, and a gamma distribution to model positive catch rates. We did not include any density covariates in estimation of age composition for consistency with models used in the previous assessments and due to computational limitations.

We estimated the spatial range assuming geometric isotropy (i.e, spatial autocorrelation declines with equal rates over distance in all cardinal directions). We used the same extrapolation grid as implemented for abundance indices, but here we modeled spatial and spatiotemporal fields with a mesh with a coarser spatial resolution than the index model, here using 50 “knots”. This reduction in the spatial resolution of the model, relative to that used for the abundance indices, was necessary due to the increased computational load of fitting multiple age categories.

The estimates of abundance at age were bias-corrected with the same method as the abundance index. We implemented the same diagnostics to check convergence and model fit as those used for the abundance index. Results compared favorably among methods (Figure 34).

Both the model-based bottom trawl index and age composition estimates were calculated using code in the AFSC GAP model-based indices GitHub repository (<https://github.com/afsc-gap-products/model-based-indices>). Design-based estimates of bottom trawl products were calculated using code in the `gapindex` R package (<https://github.com/afsc-gap-products/gapindex>).

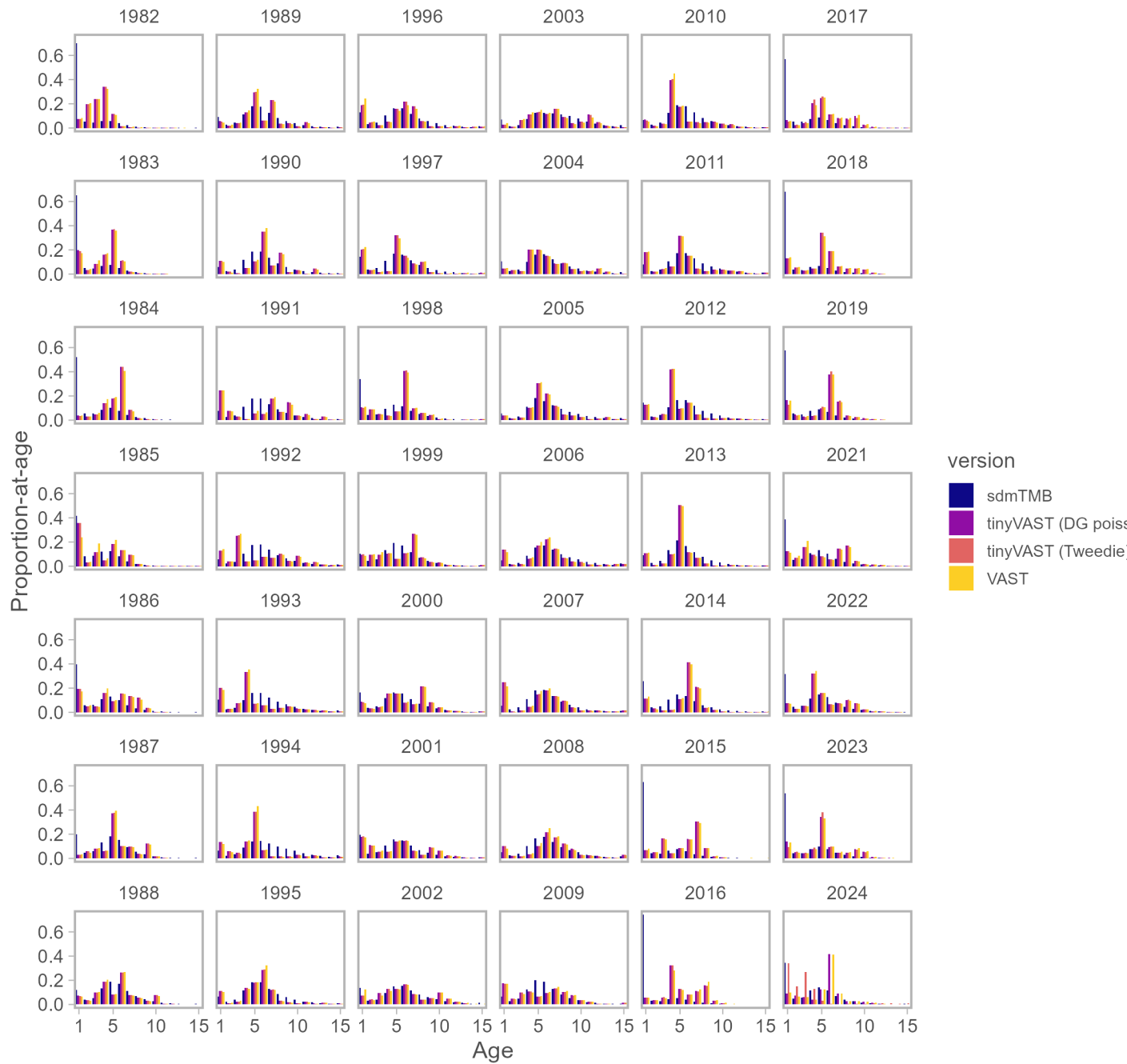


Figure 34: Model results comparing last year's model with the revised sdmTMB model.

References

- Adams, Grant D., Kirstin K. Holsman, Steven J. Barbeaux, Martin W. Dorn, James N. Ianelli, Ingrid Spies, Ian J. Stewart, and André E. Punt. 2022. “An Ensemble Approach to Understand Predation Mortality for Groundfish in the Gulf of Alaska.” *Fisheries Research* 251: 106303. <https://doi.org/https://doi.org/10.1016/j.fishres.2022.106303>.
- Anderson, S. C., E. J. Ward, P. A. English, and L. A. K. Barnett. 2022. “sdmTMB: An R Package for Fast, Flexible, and User-Friendly Generalized Linear Mixed Effects Models with Spatial and Spatiotemporal Random Fields.” *BioRxiv*, 2022–03. <https://doi.org/10.1101/2022.03.24.485545>.
- Cheng, Matthew. 2025. *SPoCK: Spatial (s) Population (Po) Model with Connectivity (c) Across Generalized Komponenten (k)*.
- Cheng, Matthew L. H., Daniel R. Goethel, Peter-John F. Hulson, Michael J. Wilberg, Craig Marsh, and Curry J. Cunningham. 2025. “Misspecifying Sex-Structured Dynamics in Stock Assessment Models.” *Fish and Fisheries*, February, 1–19. <https://doi.org/10.1111/faf.12891>.
- Dunn, K. P., and G. K. Smyth. 1996. “Randomized Quantile Residuals.” *Journal of Computational and Graphical Statistics* 5 (1): 1–10. <https://doi.org/10.1080/10618600.1996.10474708>.
- Grandin, C. J., K. F. Johnson, A. M. Edwards, and A. M. Berger. 2024. “Status of the Pacific Hake (Whiting) Stock in U.S. And Canadian Waters in 2024.” Joint Technical Committee of the U.S.; Canada Pacific Hake/Whiting Agreement, National Marine Fisheries Service; Fisheries; Oceans Canada.
- Hartig, Florian. 2021. *DHARMA: Residual Diagnostics for Hierarchical (Multi-Level / Mixed) Regression Models*. <http://florianhartig.github.io/DHARMA/>.
- Helser, Thomas E., Irina Benson, Jason Erickson, Jordan Healy, Craig Kestelle, and Jonathan A. Short. 2019. “A Transformative Approach to Ageing Fish Otoliths Using Fourier Transform Near Infrared Spectroscopy: A Case Study of Eastern Bering Sea Walleye Pollock (*Gadus Chalcogrammus*).” *Canadian Journal of Fisheries and Aquatic Sciences* 76 (5): 780–89. <https://doi.org/10.1139/cjfas-2018-0112>.
- Holsman, K. K., J. Ianelli, K. Shotwell, S. Barbeaux, K. Aydin, G. Adams, K. Kearney, A. Sulc, and S. Wassermann. 2024. “Climate-Enhanced Multispecies Stock Assessment for Walleye Pollock, Pacific Cod, and Arrowtooth Flounder in the Eastern Bering Sea.” Technical Report. Anchorage, AK: North Pacific Fishery Management Council. <https://www.npfmc.org/wp-content/PDFdocuments/SAFE/2024/EBSmultispp.pdf>.
- Ianelli, Jim. 2025. *Ebswp: Facilitates Model Running for EBS Pollock*. <http://afsc-assessments.github.io/EBSpollock/>.

- Jurado-Molina, J., P. A. Livingston, and J. N. Ianelli. 2005. “Incorporating Predation Interactions to a Statistical Catch-at-Age Model for a Predator-Prey System in the Eastern Bering Sea.” *Canadian Journal of Fisheries and Aquatic Sciences* 62 (8): 1865–73.
- Methot, Richard D., and Chantell R. Wetzel. 2013. “Stock Synthesis: A Biological and Statistical Framework for Fish Stock Assessment and Fishery Management.” *Fisheries Research* 142 (May): 86–99. <https://doi.org/10.1016/j.fishres.2012.10.012>.
- Nielsen, Anders, and Casper W. Berg. 2014. “Estimation of Time-Varying Selectivity in Stock Assessments Using State-Space Models.” *Fisheries Research* 158: 96–101. <https://doi.org/10.1016/j.fishres.2014.01.014>.
- Nielsen, Anders, Niels T Hintzen, Henrik Mosegaard, Vanessa Trijoulet, and Casper W Berg. 2021. “Multi-Fleet State-Space Assessment Model Strengthens Confidence in Single-Fleet SAM and Provides Fleet-Specific Forecast Options.” *ICES Journal of Marine Science* 78 (6): 2043–52. <https://doi.org/10.1093/icesjms/fsab078>.
- O’Leary, C. A., J. T. Thorson, J. N. Ianelli, and S. Kotwicki. 2020. “Adapting to Climate-driven Distribution Shifts Using Model-based Indices and Age Composition from Multiple Surveys in the Walleye Pollock (*Gadus Chalcogrammus*) Stock Assessment.” *Fisheries Oceanography* 29 (6): 541–57. <https://doi.org/10.1111/fog.12494>.
- Punt, A. E., D. C. Smith, K. KrusicGolub, and S. Robertson. 2008. “Quantifying Age-Reading Error for Use in Fisheries Stock Assessments, with Application to Species in Australia’s Southern and Eastern Scalefish and Shark Fishery.” *Can. J. Fish. Aquat. Sci.* 65: 1991–2005.
- Rohan, S. K., L. A. K. Barnett, and N. Charriere. 2022. “Evaluating Approaches to Estimating Mean Temperatures and Cold Pool Area from AFSC Bottom Trawl Surveys of the Eastern Bering Sea.” Tech. Mem. NMFS-AFSC-456. U.S. Dep. Commer., NOAA. <https://doi.org/10.25923/1wwh-q418>.
- Stock, Brian C., and Timothy J. Miller. 2021. “The Woods Hole Assessment Model (WHAM): A General State-Space Assessment Framework That Incorporates Time- and Age-Varying Processes via Random Effects and Links to Environmental Covariates.” *Fisheries Research* 240 (August): 105967. <https://doi.org/10.1016/j.fishres.2021.105967>.
- Thorson, J. T. 2018. “Three Problems with the Conventional Delta-Model for Biomass Sampling Data, and a Computationally Efficient Alternative.” *Canadian Journal of Fisheries and Aquatic Sciences* 75 (9): 1369–82. <https://doi.org/10.1139/cjfas-2017-0266>.
- . 2019. “Measuring the Impact of Oceanographic Indices on Species Distribution Shifts: The Spatially Varying Effect of Cold-pool Extent in the Eastern Bering Sea.” *Limnology and Oceanography* 64 (6): 2632–45. <https://doi.org/10.1002/lno.11238>.
- Thorson, J. T., S. C. Anderson, P. Goddard, and C. N. Rooper. 2024. “tinyVAST: R Package with an Expressive Interface to Specify Lagged and Simultaneous Effects in Multivariate Spatio-Temporal Models.” *arXiv Preprint* arXiv:2401.10193.
- Thorson, J. T., and K. Kristensen. 2016. “Implementing a Generic Method for Bias Correction in Statistical Models Using Random Effects, with Spatial and Population Dynamics Examples.” *Fisheries Research* 175: 66–74. <https://doi.org/10.1016/j.fishres.2015.11.016>.